Fast & Resource-Efficient Toxic-Text Classifiers

Bilingual (RU + EN) safety filters: quality, speed, robustness

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Generative AI · Spring 2026

Problem

Toxic-text filters sit on latency-sensitive paths. Three axes in tension:

Quality
Catch subtle, context-dependent toxicity. Miss rate directly harms users.

Speed
Sub-millisecond latency for real-time filtering. CPU-only budgets.

Robustness
Hold up against l33tspeak, character spacing, Cyrillic↔Latin homoglyphs.

Question: how do classical, full fine-tune, and LoRA compare on the same bilingual corpus?

Data

460K	4	92K	2
samples (after LSH dedup)	public datasets merged	test set (stratified, seed=42)	languages (RU + EN)

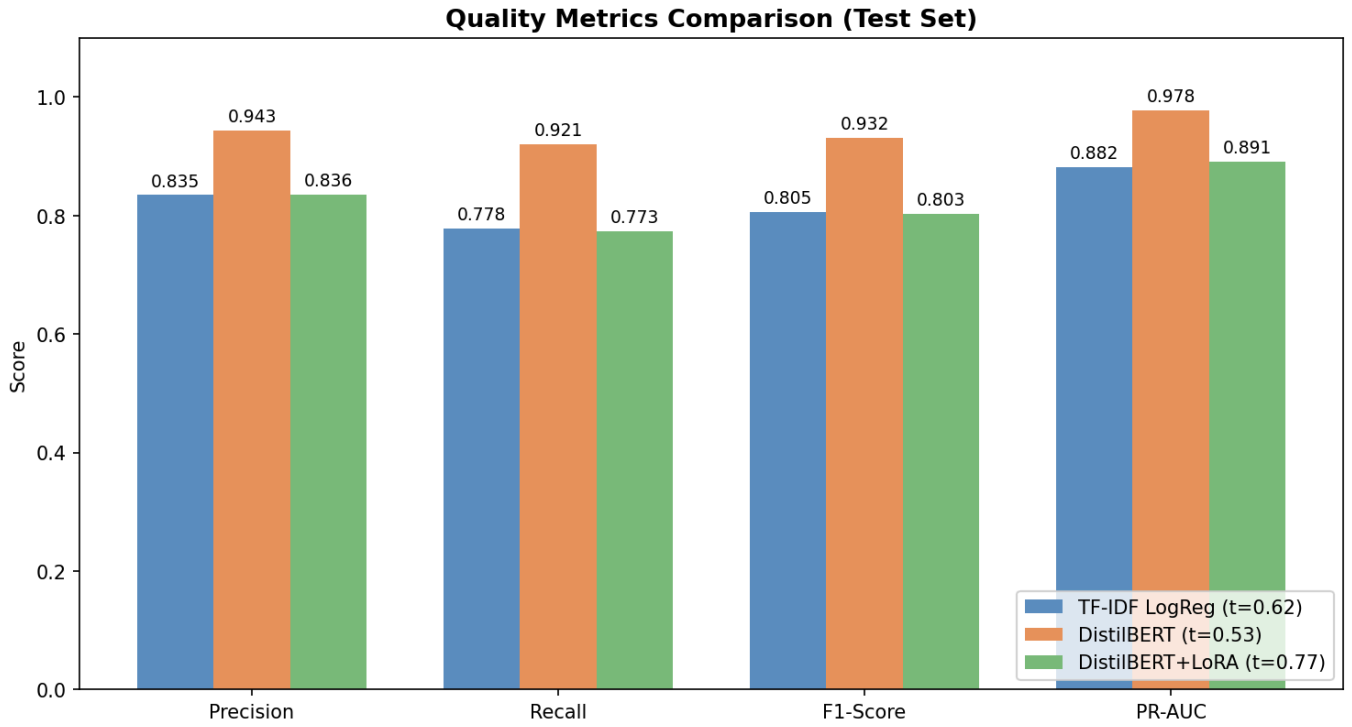
Cleaning stripped HTML/URLs/IPs/Wikipedia metadata. MinHash-LSH (t=0.80) removed 2.1% near-duplicates. Obfuscated stress set built for robustness eval.

Approach: three architectures, one protocol

Model	Params	Training data	Notes
TF-IDF + LogReg	10K features	133K (balanced)	unigrams + bigrams, L2
DistilBERT (full FT)	67.6M	331K	`distilbert-base-uncased`, 3 ep
DistilBERT + LoRA	**739K (1.09%)**	90K (balanced)	r=8, α=16, q_lin + v_lin

Unified eval: same 92K test split. Thresholds tuned on val (sweep 0.10–0.90, argmax F1).

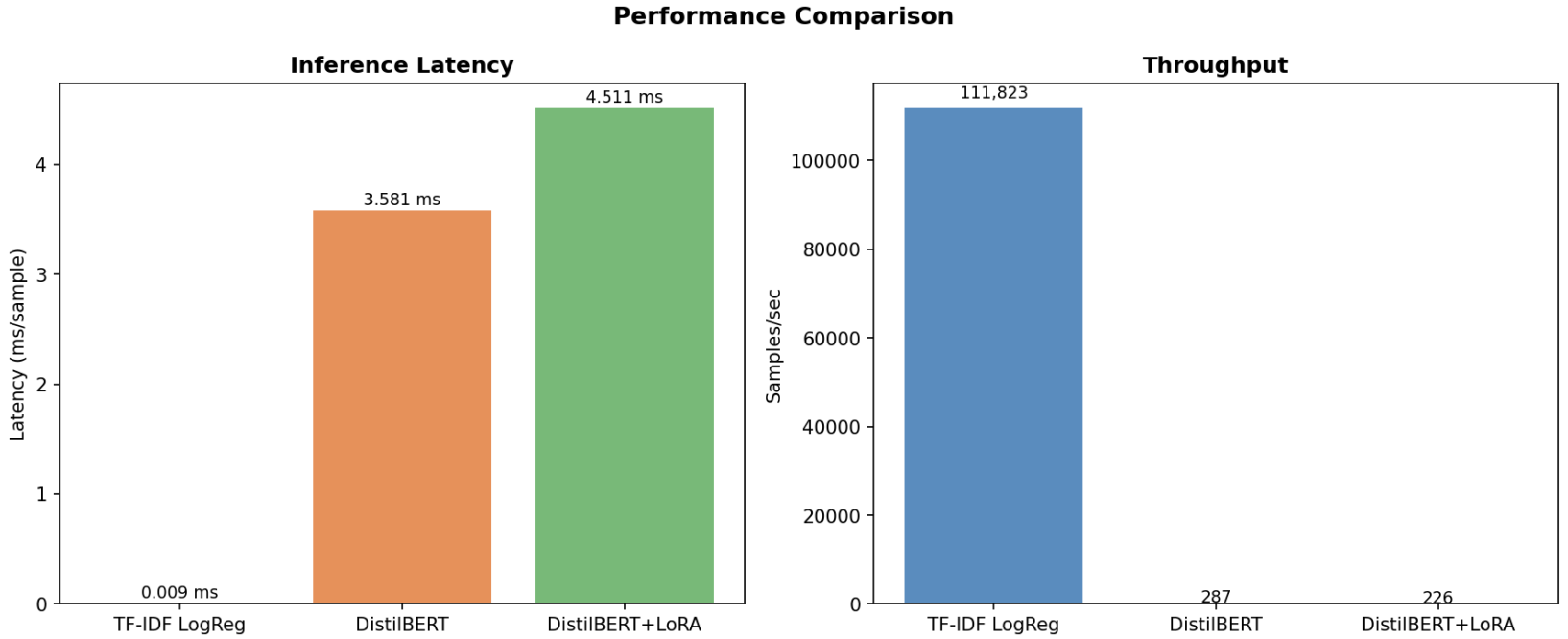
Headline results



Model	F1	Latency	Throughput
TF-IDF + LogReg	0.8054	**0.009 ms**	**112K/s**
DistilBERT	**0.9318**	3.58 ms	287/s
DistilBERT + LoRA	0.8035	4.51 ms	226/s

DistilBERT **+12.6pp** F1 over LogReg. LogReg **~390×** throughput of transformers.

Speed vs quality tradeoff



LoRA stuck in the **worst quadrant**: transformer latency, LogReg-level quality. Training efficiency only.

Per-language: Russian vs English

Model	EN F1	RU F1	Gap
TF-IDF + LogReg	0.8278	0.7851	**+4.3pp**
DistilBERT (full FT)	0.9364	0.9278	**+0.9pp**
DistilBERT + LoRA	0.8491	0.7608	**+8.8pp**

`distilbert-base-uncased` is English-pretrained, lowercases, strips accents. Cyrillic → byte-level WordPiece fallback.

Full fine-tuning absorbs the mismatch. LoRA cannot. 1% trainable params insufficient to realign Russian representation.

Ablations: overview

Four ablation studies, all evaluated under the same protocol as the main results:

#	Question	Models	Headline
1	Does balancing classes help?	LogReg / DistilBERT / LoRA	model-dependent
2	Does class-weighted loss help DistilBERT?	DistilBERT	**−3.8pp** F1
3	Is the LoRA gap from data size or architecture?	LoRA vs full FT @ 90K	**+7.1pp** full FT
4	Robustness under obfuscation + deobfuscation defense	all 3	**76%** recovery

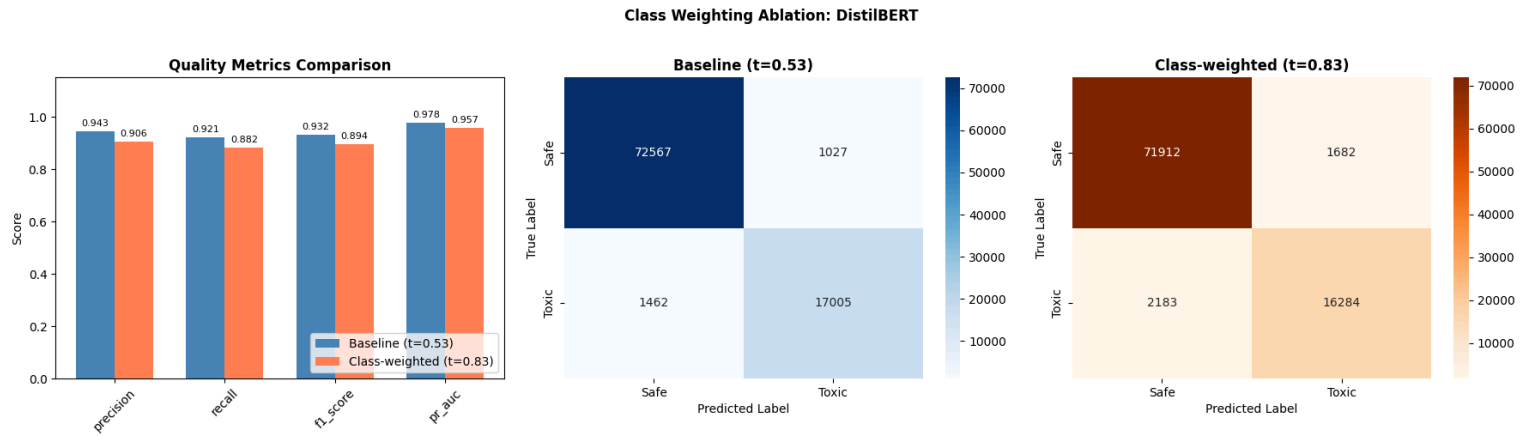
Each ablation derives modified training data from the global split. Val/test held fixed. Promoted weights: balanced LogReg (133K), balanced same-size LoRA (90K).

Ablation 1: class imbalance (undersampling)

Model	Baseline F1	Balanced F1	Δ	Verdict
LogReg (331K → **133K balanced**)	0.7647	**0.8054**	**+4.1pp**	helps · promoted
DistilBERT (331K → 133K balanced)	0.9318	0.8844	–4.7pp	hurts (data loss)
LoRA (90K → 36K balanced, naive)	0.7929	0.7300	–6.3pp	hurts (data loss)
LoRA (90K → **90K balanced**, same-size)	0.7929	**0.8035**	**+1.1pp**	helps · promoted

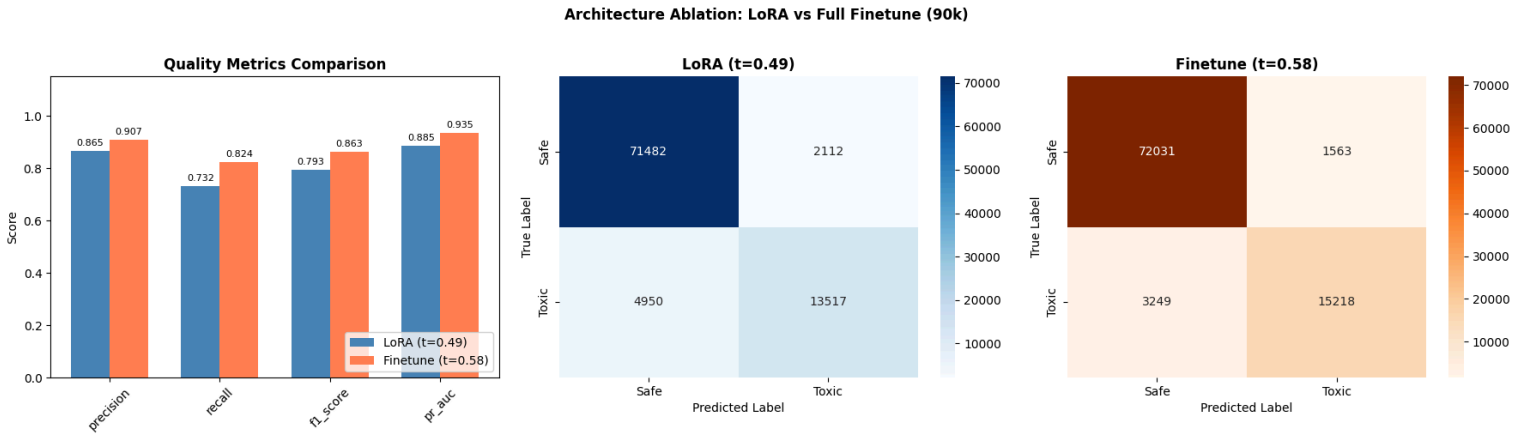
Pattern is model-dependent. Linear models gain from balanced priors. Pretrained transformers on 331K already have no majority-class bias to correct → undersampling only loses data. Holding LoRA's training size constant isolates the distribution effect from data loss.

Ablation 2: class weighting hurts full FT



DistilBERT @ 331K with 4× weight on toxic class: F1 **–3.8pp** (0.9318 → 0.8939) · PR-AUC **–2.1pp** · threshold drifts **0.53 → 0.83**. No majority-class bias to correct → gradient distortion breaks calibration. Both undersampling (–4.7pp) and weighting (–3.8pp) assume imbalance hurts; for full FT on 331K that assumption is wrong.

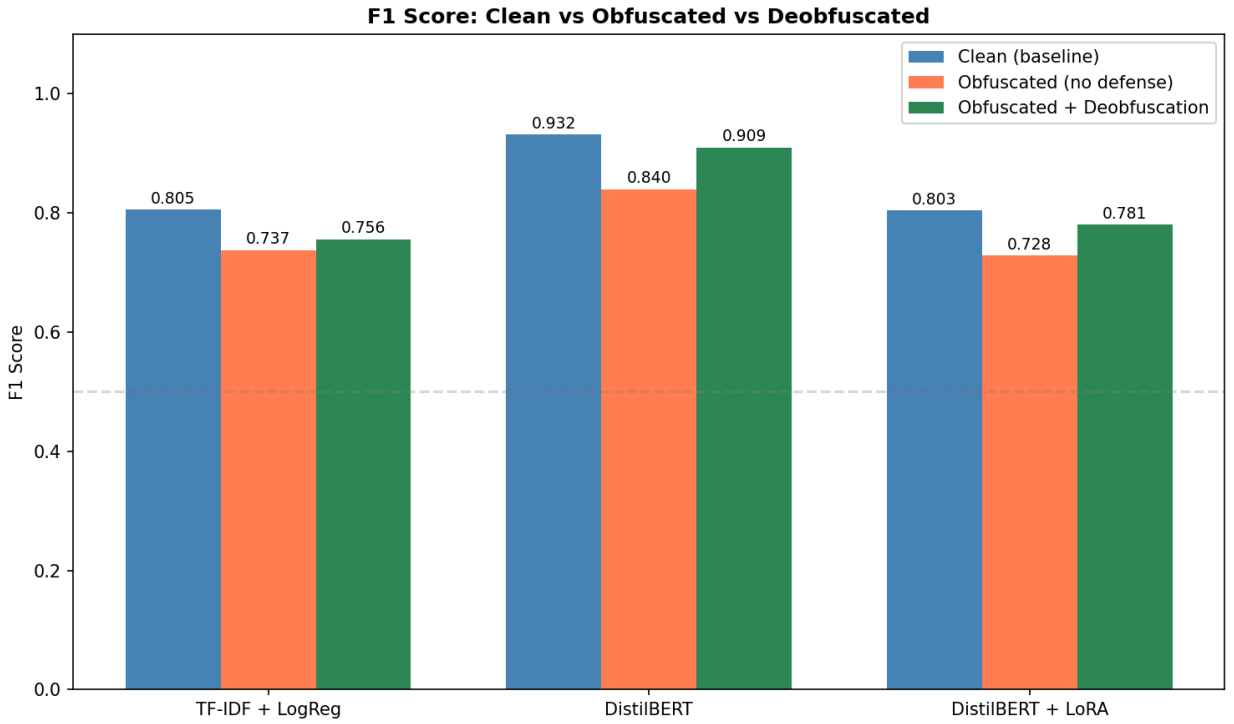
Ablation 3: LoRA vs full FT on same 90K



Model	Precision	Recall	F1	PR-AUC
LoRA (90K)	0.8649	0.7320	0.7929	0.8853
Full FT (90K)	**0.9069**	**0.8241**	**0.8635**	**0.9354**
Δ	+0.042	**+0.092**	**+7.1pp**	+0.050

Total LoRA vs full FT gap on the main split is **13.9pp**. Decomposes: **~7pp architecture, ~7pp data size**. LoRA's 1% trainable budget limits discriminative features for the toxic class. Recall hit hardest (+9.2pp).

Ablation 4: robustness under obfuscation



Model	Clean	Obfuscated	+ Deobf.	Recovery
LogReg	0.8054	0.7373	0.7555	26.8%
DistilBERT	**0.9318**	0.8395	**0.9095**	**75.8%**
LoRA	0.8035	0.7283	0.7806	69.5%

Rule-based deobfuscation preprocessor: **zero retraining, ~76% recovery on transformers.**

Deployment recommendations

Use case	Pick	Why
Real-time, CPU-only	**TF-IDF + LogReg**	0.009 ms, 0.5 MB, F1=0.81
Batch safety review	**DistilBERT + deobf.**	F1=0.93 clean / 0.91 obf; smallest EN/RU gap
Rapid prototyping	LoRA	6× faster training, **not** for prod serving

Every transformer deployment → add the deobfuscation preprocessor. Free, effective.

Limitations

- **Domain mismatch.** Social-media comments only. LLM-output moderation unverified.
- **English-centric backbone.** XLM-R / mDeBERTa would avoid the EN/RU gap by design.
- **Single-seed training.** No variance across training seeds; small deltas indicative only.
- **Adversarial scope.** 4 rule-based obfuscation families. No paraphrase or semantic attacks.
- **Binary label collapse.** Severity, target, context discarded across 4 source datasets.
- **Rule-based deobfuscation.** Brittle to novel evasion; learned normalizer is future work.

Takeaways

1. Full fine-tuning dominates on quality (F1 0.93) *and* closes the RU/EN gap.

2. LoRA isn't a drop-in substitute: worst-quadrant latency/quality, largest language gap.

3. A free preprocessor recovers 76% of obfuscation loss on transformers.

Repo: `github.com/ArthurBabkin/GenAI-Safety-Fliter` · full report in `docs/final/final.md`

Team contributions

Arthur Babkin

- Collected data, prepared dataset
- Baseline TF-IDF implementation
- Ablation 2 (class weighting)
- Ablation 3 (LoRA vs full FT @ 90K)

Alexander Malyy

- Baseline DistilBERT + LoRA implementation
- Code refactor, unified methodology
- Ablation 1 (class imbalance)
- Ablation 4 (robustness + deobfuscation)